



11318 - How do novae form dust, and how much dust survives?

Cycle: 5, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Peter Craig (PI)	Michigan State University
Dr. Laura Chomiuk (CoI)	Michigan State University
Elias Aydi (CoI)	Michigan State University
Prof. David J. Sand (CoI)	University of Arizona
Prof. Jay Strader (CoI)	Michigan State University

OBSERVATIONS

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V5668 Observations				
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	2	V5668 Background	MIRI Medium Resolution Spectroscopy	(2) V5668-Sgr-Background
V339 Del Observations				
	3	V339 Del MRS Main	MIRI Medium Resolution Spectroscopy	(3) V339-Del
	4	V339 Del Background	MIRI Medium Resolution Spectroscopy	(4) V339-Del-Background
V445 Pup Observations				
	5	V445 Pup MRS Main	MIRI Medium Resolution Spectroscopy	(5) V445-Pup
	6	V445 Pup Background	MIRI Medium Resolution Spectroscopy	(7) V445-Pup-Background
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V1280 Sco Observations				
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ABSTRACT

We propose JWST/MIRI MRS observations of four Galactic novae that are 9–24 years old, enabling us to resolve the nova ejecta, image surviving dust, and place it in context of the 10^4 K warm ejecta. These observations will test how dust is formed in novae, differentiating between formation in ballistic clumps and formation at radiative shock fronts. These observations will also measure how much dust survives the harsh radiation environment of the nova eruption, with the ability to detect and image the dust even if just 0.1% of it survives to late times. Novae present a rare opportunity amongst astrophysical transients for JWST to produce stunning images of the multi-phase ejecta, and yet, this program would be the first time JWST has observed any nova. By developing an understanding of dust formation and survival in nova eruptions, we will anchor our understanding of dust production in other distant astrophysical transients like supernovae, which may in turn contribute significantly to the cosmic dust budget.

OBSERVING DESCRIPTION

We intend to observe 4 nova remnants with MIRI MRS, complete with background exposures. Each MIRI MRS exposure will use the long wavelength sub-channel (C) of all 4 MRS channels, and be part of a non-interruptable sequence with a related background exposure. The 4 novae targeted here are V339 Del, V5668 Sgr, V1280 Sco and V445 Pup. In the case of V445 Pup, we also intend to obtain MIRI Imaging in the F1000W, F1500W and F2100W to make use of the larger field of view, with no special requirements for the MIRI imaging.

Proposal 11318 - Targets - How do novae form dust, and how much dust survives?

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	V5668-Sgr	RA: 18 36 56.8279 (279.2367829d) Dec: -28 55 39.98 (-28.92777d) Equinox: J2000	Proper Motion RA: -10.6 mas/yr Proper Motion Dec: -10.400000041954627 mas/yr Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Novae] Extended=YES				
	(2)	V5668-Sgr-Background	RA: 18 36 57.6196 (279.2400817d) Dec: -28 55 52.49 (-28.93125d) Equinox: J2000	Proper Motion RA: -10.6 mas/yr Proper Motion Dec: -10.400000041954627 mas/yr Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Calibration Description=[Telescope/sky background] Extended=YES				
	(3)	V339-Del	RA: 20 23 30.6865 (305.8778604d) Dec: +20 46 3.78 (20.76772d) Equinox: J2000	Proper Motion RA: -4.918 mas/yr Proper Motion Dec: -7.820999962859787 mas/yr Parallax: 3.814E-4" Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Novae] Extended=YES					
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Calibration Description=[Telescope/sky background] Extended=NO					
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Proper motion estimates are from : http://www.cbat.eps.harvard.edu/iauc/07500/07556.html The coordinates for this target have been shifted by 4", in order to select one half of the bipolar outflow within the MIRI MRS Channels. Category=Star Description=[Novae] Extended=YES					

Proposal 11318 - Targets - How do novae form dust, and how much dust survives?

(6)	V445-Pup-TA	RA: 07 37 58.3506 (114.4931275d) Dec: -25 56 46.23 (-25.94617d) Equinox: J2000	Proper Motion RA: -1.259 mas/yr Proper Motion Dec: 0.732 mas/yr Parallax: 2.015000000000002E-4" Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. The parameters for this source (for spectral type and ETC Calculations) are estimated based on Gaia DR3 stellar parameters, providing an effective temperature of 5128 K with alogg of 2.84 cm/s^2 Category=Star Description=[G stars]			
(7)	V445-Pup-Background	RA: 07 37 51.1430 (114.4630958d) Dec: -25 57 12.90 (-25.95358d) Equinox: J2000	Epoch of Position: 2000
Comments: Background observation location for V445 Pup. Aladin shows no sources from 2Mass or Wise nearby Category=Calibration Description=[Telescope/sky background] Extended=NO			
(8)	V1280-Sco	RA: 16 57 41.2186 (254.4217442d) Dec: -32 20 35.70 (-32.34325d) Equinox: J2000	Proper Motion RA: -0.1150000000000002 mas/yr Proper Motion Dec: -1.3920000128564425 mas/yr Parallax: 2.747E-4" Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Novae] Extended=YES			
(9)	V1280-Sco-Background	RA: 16 57 50.2492 (254.4593717d) Dec: -32 20 57.60 (-32.34933d) Equinox: J2000	Proper Motion RA: -0.1150000000000002 mas/yr Proper Motion Dec: -1.3920000128564425 mas/yr Parallax: 2.747E-4" Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Calibration Description=[Telescope/sky background] Extended=YES			

Proposal 11318 - Observation 1 - How do novae form dust, and how much dust survives?

Observation	Proposal 11318, Observation 1: V5668 Sgr MRS Main												Sat May 16 00:00:11 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(1)	V5668-Sgr	RA: 18 36 56.8279 (279.2367829d) Dec: -28 55 39.98 (-28.92777d) Equinox: J2000				Proper Motion RA: -10.6 mas/yr Proper Motion Dec: -10.400000041954627 mas/yr Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[Novae] Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray		Grating Wheel Direction	
	F560W	All MRS				YES				FULL		Allow Auto Reorder	
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F560W	FASTR1	100	2	1	Dither 1	4	8	2231.132	
	1	LONG(C)	MRSLONG		FASTR1	100	2	1	Dither 1	4	8	2231.132	227692.7
	1	LONG(C)	MRSSHORT		FASTR1	100	2	1	Dither 1	4	8	2231.132	227692.6

Proposal 11318 - Observation 1 - How do novae form dust, and how much dust survives?

Special Requirements	Group Observations 1, 2, Non-interruptible
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Proposal 11318 - Observation 2 - How do novae form dust, and how much dust survives?

Observation	Proposal 11318, Observation 2: V5668 Background												Sat May 16 00:00:11 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(2)	V5668-Sgr-Background	RA: 18 36 57.6196 (279.2400817d) Dec: -28 55 52.49 (-28.93125d) Equinox: J2000				Proper Motion RA: -10.6 mas/yr Proper Motion Dec: -10.400000041954627 mas/yr Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Calibration Description=[Telescope/sky background] Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray		Grating Wheel Direction	
		All MRS				YES				FULL		Allow Auto Reorder	
Dithers	#	Dither Type				Optimized For				Direction			
	1	2-Point				EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F560W	FASTR1	100	2	1	Dither 1	2	4	1115.566	
	1	LONG(C)	MRSLONG		FASTR1	100	2	1	Dither 1	2	4	1115.566	
	1	LONG(C)	MRSSHORT		FASTR1	100	2	1	Dither 1	2	4	1115.566	

Proposal 11318 - Observation 2 - How do novae form dust, and how much dust survives?

Special Requirements	Group Observations 1, 2, Non-interruptible
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Proposal 11318 - Observation 3 - How do novae form dust, and how much dust survives?

Observation	Proposal 11318, Observation 3: V339 Del MRS Main												Sat May 16 00:00:11 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(3)	V339-Del	RA: 20 23 30.6865 (305.8778604d) Dec: +20 46 3.78 (20.76772d) Equinox: J2000				Proper Motion RA: -4.918 mas/yr Proper Motion Dec: -7.820999962859787 mas/yr Parallax: 3.814E-4" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[Novae] Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
	F560W	All MRS				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F560W	FASTR1	100	2	1	Dither 1	4	8	2231.132	
	1	LONG(C)	MRSLONG		FASTR1	100	2	1	Dither 1	4	8	2231.132	227692.3
	1	LONG(C)	MRSSHORT		FASTR1	100	2	1	Dither 1	4	8	2231.132	227692.2

Proposal 11318 - Observation 3 - How do novae form dust, and how much dust survives?

Special Requirements	Group Observations 3, 4, Non-interruptible
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Proposal 11318 - Observation 4 - How do novae form dust, and how much dust survives?

Observation	Proposal 11318, Observation 4: V339 Del Background												Sat May 16 00:00:11 GMT 2026
	Diagnostic Status: Warning Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(4)	V339-Del-Background	RA: 20 23 29.0129 (305.8708871d) Dec: +20 46 7.75 (20.76882d) Equinox: J2000				Parallax: 3.814E-4" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Calibration Description=[Telescope/sky background] Extended=NO												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
		All MRS				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	2-Point				EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F560W	FASTR1	100	2	1	Dither 1	2	4	1115.566	
	1	LONG(C)	MRSLONG		FASTR1	100	2	1	Dither 1	2	4	1115.566	
	1	LONG(C)	MRSSHORT		FASTR1	100	2	1	Dither 1	2	4	1115.566	

Proposal 11318 - Observation 4 - How do novae form dust, and how much dust survives?

Special Requirements	Group Observations 3, 4, Non-interruptible
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Proposal 11318 - Observation 5 - How do novae form dust, and how much dust survives?

Observation	Proposal 11318, Observation 5: V445 Pup MRS Main												Sat May 16 00:00:11 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(5)	V445-Pup	RA: 07 37 57.0172 (114.4875717d)				Proper Motion RA: 0.0047 arcsec/yr						
			Dec: -25 56 58.01 (-25.94945d)				Proper Motion Dec: 0.0064 arcsec/yr						
			Equinox: J2000				Epoch of Position: 2000						
		Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
		Proper motion estiamtes are from : http://www.cbat.eps.harvard.edu/iauc/07500/07556.html											
Acquisition	The coordinates for this target have been shifted by 4", in order to select one half of the bipolar outflow within the MIRI MRS Channels.												
	Category=Star												
	Description=[Novae]												
	Extended=YES												
	#	Target											
1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
	F560W	All MRS				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F560W	FASTR1	50	2	1	Dither 1	4	8	1121.116	
	1	LONG(C)	MRSLONG		FASTR1	50	2	1	Dither 1	4	8	1121.116	227692.15
	1	LONG(C)	MRSSHORT		FASTR1	50	2	1	Dither 1	4	8	1121.116	227692.14

Proposal 11318 - Observation 5 - How do novae form dust, and how much dust survives?

Special Requirements	Group Observations 5, 6, Non-interruptible
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Proposal 11318 - Observation 6 - How do novae form dust, and how much dust survives?

Observation	Proposal 11318, Observation 6: V445 Pup Background												Sat May 16 00:00:11 GMT 2026
	Diagnostic Status: Warning												
Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(7)	V445-Pup-Background	RA: 07 37 51.1430 (114.4630958d)					Epoch of Position: 2000					
			Dec: -25 57 12.90 (-25.95358d)										
			Equinox: J2000										
			Comments: Background observation location for V445 Pup. Aladin shows no sources from 2Mass or Wise nearby										
Acquisition	Category=Calibration												
	Description=[Telescope/sky background]												
	Extended=NO												
	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
		All MRS				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For			Direction			
	1	2-Point					EXTENDED SOURCE			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F560W	FASTR1	50	2	1	Dither 1	2	4	560.558	
	1	LONG(C)	MRSLONG		FASTR1	50	2	1	Dither 1	2	4	560.558	
	1	LONG(C)	MRSSHORT		FASTR1	50	2	1	Dither 1	2	4	560.558	

Proposal 11318 - Observation 6 - How do novae form dust, and how much dust survives?

Special Requirements	Group Observations 5, 6, Non-interruptible
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Proposal 11318 - Observation 9 - How do novae form dust, and how much dust survives?

Observation	Proposal 11318, Observation 9: V445 Pup MIRI Imaging										Sat May 16 00:00:11 GMT 2026
	Diagnostic Status: Warning										
Observing Template: MIRI Imaging											
Diagnostics	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(5)	V445-Pup	RA: 07 37 57.0172 (114.4875717d)			Proper Motion RA: 0.0047 arcsec/yr					
			Dec: -25 56 58.01 (-25.94945d)			Proper Motion Dec: 0.0064 arcsec/yr					
			Equinox: J2000			Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	Proper motion estiamtes are from : http://www.cbat.eps.harvard.edu/iauc/07500/07556.html										
	The coordinates for this target have been shifted by 4", in order to select one half of the bipolar outflow within the MIRI MRS Channels.										
Template	Category=Star										
	Description=[Novae]										
Dithers	Extended=YES										
	Subarray										
Dithers	FULL										
	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	4						LARGE	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F1000W	FASTR1	5	1	1	Dither 1	4	4	55.501	227692.17
	2	F1500W	FASTR1	5	1	1	Dither 1	4	4	55.501	227692.18
	3	F2100W	FASTR1	5	1	1	Dither 1	4	4	55.501	227692.19

Proposal 11318 - Observation 7 - How do novae form dust, and how much dust survives?

Observation	Proposal 11318, Observation 7: V1280 Sco Main MRS												Sat May 16 00:00:11 GMT 2026
	Diagnostic Status: Warning												
Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(8)	V1280-Sco	RA: 16 57 41.2186 (254.4217442d) Dec: -32 20 35.70 (-32.34325d) Equinox: J2000					Proper Motion RA: -0.11500000000000002 mas/yr Proper Motion Dec: -1.3920000128564425 mas/yr Parallax: 2.747E-4" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[Novae] Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
	F560W	All MRS				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For			Direction			
	1	4-Point					EXTENDED SOURCE			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F560W	FASTR1	100	2	1	Dither 1	4	8	2231.132	
	1	LONG(C)	MRSLONG		FASTR1	100	2	1	Dither 1	4	8	2231.132	227692.11
	1	LONG(C)	MRSSHORT		FASTR1	100	2	1	Dither 1	4	8	2231.132	227692.10

Proposal 11318 - Observation 7 - How do novae form dust, and how much dust survives?

Special Requirements	Group Observations 7, 8, Non-interruptible
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Proposal 11318 - Observation 8 - How do novae form dust, and how much dust survives?

Observation	Proposal 11318, Observation 8: V1280 Sco Background												Sat May 16 00:00:11 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(9)	V1280-Sco-Background	RA: 16 57 50.2492 (254.4593717d) Dec: -32 20 57.60 (-32.34933d) Equinox: J2000					Proper Motion RA: -0.11500000000000002 mas/yr Proper Motion Dec: -1.3920000128564425 mas/yr Parallax: 2.747E-4" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Calibration												
	Description=[Telescope/sky background] Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
		All MRS				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For			Direction			
	1	2-Point					EXTENDED SOURCE			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F560W	FASTR1	100	2	1	Dither 1	2	4	1115.566	
	1	LONG(C)	MRSLONG		FASTR1	100	2	1	Dither 1	2	4	1115.566	
	1	LONG(C)	MRSSHORT		FASTR1	100	2	1	Dither 1	2	4	1115.566	

Proposal 11318 - Observation 8 - How do novae form dust, and how much dust survives?

Special Requirements	Group Observations 7, 8, Non-interruptible
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